Atom combining

Ions: Positively or negatively charged particles. It is charged because it has unequal number of protons and electrons.

Positive ions: Cations, Groups 1, 2, 3, transition metals, Hydrogen

↓

lose electrons

Negative ions: Anions, Groups 5, 6, 7

↓

gain electrons

Ionic bond: The bond of attraction that forms between the positively charged and negatively charged ions.

Ionic lattice structure: A regular pattern of alternating positive and negative ions held together by strong ionic bonds.

* A metal reacts with a non-metal to form an ionic bond, where the metal loses electrons, and the non-metal gains them. The compound will have an overall charge of zero.

Types of hydrogen:

* Atomic hydrogen: Consists of one proton and one unpaired electron

* Molecular hydrogen: Two protons and two electrons. Most common type of hydrogen because it is stable with neutral charge.

* Positive hydrogen: H^+: Proton: loses an electron to be left with a proton

* Negative hydrogen: H^-: Hydride: Gains an electron to be negatively charged.

Ions

Monoatomic

Cations:

Groups 1
2
3
+
transition metals
+
Hydrogen

Anions:

Groups 5
6
7

Polyatomic

Cations:

* Ammonium: NH_4^+

Anions:

* Nitrite: NO_2^-
* Nitrate: NO_3^-
* Sulfite: SO_3^{2-}
* Sulfate: SO_4^{2-}
* Carbonate: CO_3^{2-}
* Hydrogen carbonate: HCO_3^-
* Hydroxide: OH^-

Covalent bond: When two atoms share electrons in order to get a full outer shell. Non-metal reacting with non-metal

Types of covalent bonds:

→ Single covalent bond: 1 pair of shared electrons

Halogens: F-F, Cl-Cl, Br-Br, I-I

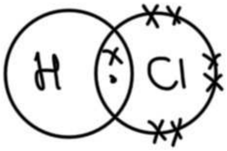
→ Double covalent bond: 2 pairs of shared electrons

Oxygen: O=O

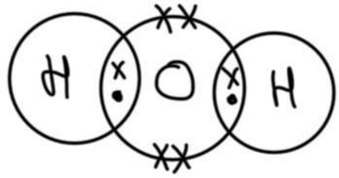
→ Triple Covalent bond: 3 pairs of shared electrons

Nitrogen: $N \equiv N$

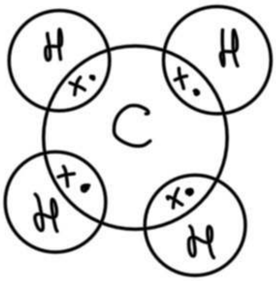
Hydrogen Chloride: HCl



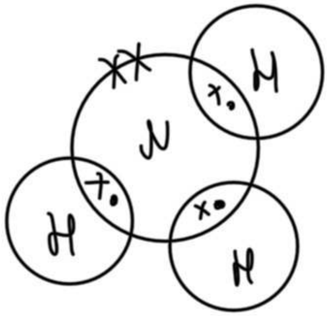
Water: H_2O



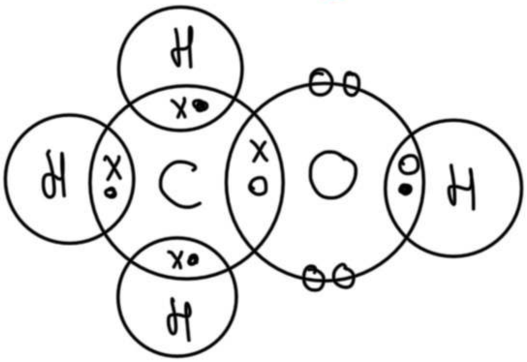
Methane: CH_4



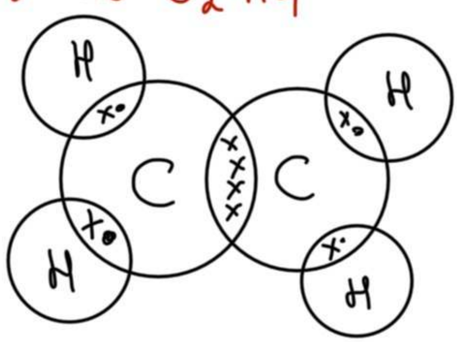
Ammonia: NH_3



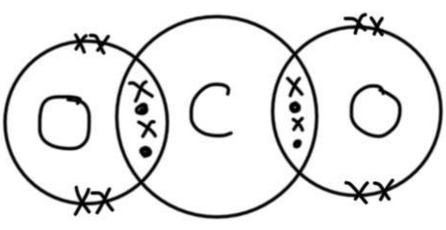
Methanol: CH_3OH



Ethene: C_2H_4



Carbon dioxide: CO_2



Properties of ionic compounds: (solids at room temperature)

* High melting and boiling points. This is because the ionic bonds are very strong, so it takes a lot of energy to break the lattice
* Soluble in water
* Conduct electricity only when molten or dissolved because they'll have free ions.

Properties of covalent compounds:

* Low melting and boiling points. This is because the attraction between the molecules is weak, so it doesn't take much energy to break the lattice. (liquids or gases at room temperature)
* Insoluble in water, but are soluble in other substances.
* Do not conduct electricity.

Giiant covalent compounds

Diamond: Made up when a carbon atom forms a covalent bond with four other carbon atoms, forming a tetrahedral shape.

Properties:

1- Very hard as each atom is held in place by for strong covalent bonds. → So, used in tools for cutting & drilling.

2- Very high melting point: 3550 °C

3- Cannot conduct electricity because there are no free ions / electrons

4- Sparkle when cut → So, used for jewellery.

Silicon (IV) oxide: Silica: Made when a silicon atom bonds covalently with four oxygen atoms. This form a tetrahedral shape around the silicon. Each oxygen atom bonds covalently with two silicon atoms. This forms a linear shape around the oxygen

Properties:

1- Hard substance, can scratch things → Used in sandpaper

2- High melting point: 1710 → Used in bricks for lining furnaces.

3- Cannot conduct electricity

4- Let's light through → used to make glass and lenses

Giant covalent bonds

Graphite: Made when each carbon atom forms a covalent bond with three other carbon atoms, forming a hexagonal ring of six carbon atoms. The rings form flat sheets that lie on top of each other, held together by weak forces

Properties:

1- Soft and slippery because the sheets can slide over each other
→ So, used as a lubricant for engines and locks.

2- Good conductor of electricity because it has free electrons →
Used for electrodes and connecting brushes in generators.

3- Has a low melting and boiling points because weak forces of attraction hold the layers together.

4- Soft and dark in color → Used in pencils when mixed with clay

Diamond and graphite are allotropes of carbon

Allotrope: Two structures composed of the same element with different structures

Metallic bond: The attraction between metal ions and the sea of delocalized electrons → Which are electrons that can move freely and are not tied to any ion.

Properties:

1- High melting points because it takes a lot of energy to break the lattice, with its strong metallic bonds.

2- Metals are malleable and ductile because layers can slide over each other.

3- Good conductors of heat & electricity because they have free electrons that transfer the heat and charge.